

SHIH-HSIN CHUANG

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Education

University of Toronto St. George Campus - Trinity College

Sept. 2022 – June 2026

Honours Bachelor of Science - Computer Science Specialist with Focus in AI & Statistics Minor

Toronto, Canada

- **CGPA:** 3.93/4.00
- **Relevant Coursework:** Machine Learning, Neural Networks, Computer Vision, Natural Language Processing, Databases, Data Structures & Algorithms, Software Design, Operating Systems, Probability & Statistics
- **Awards:** Dean's List Scholar, Chancellor's Scholarship

Technical Skills

Languages: Python, R, Java, C, SQL, GraphQL, JavaScript, TypeScript, Bash, HTML, CSS

Frameworks/Libraries: PyTorch, TensorFlow, Scikit-Learn, LangChain, NumPy, OpenCV, Keras, NLTK, Pandas, Matplotlib, React, Node.js

Tools: Git, GitHub, Linux, PostgreSQL, Prisma, Docker, Jupyter Notebook, AWS (SageMaker, S3, Lambda, SQS, SNS)

Work Experience

Software Development Engineer Intern

May 2025 – July 2025

Amazon Web Services

Vancouver, Canada

- Developed an agentic AI on-call assistant for AWS RDS using the Model Context Protocol (MCP) to automate root cause analysis, standardizing incident response workflows for multiple cross-functional engineering teams
- Achieved 93% accuracy in root cause analysis and SOP recommendations, leading to widespread adoption across the RDS department and positive feedback from diverse on-call engineers during live testing

Undergraduate Researcher – Natural Language Processing & Databases

Jan. 2025 – Present

Vector Institute, Dr. Anna Goldenberg

Toronto, Canada

- Building a RAG-based system that translates natural language queries into SQL for multimodal healthcare datasets, integrating FAISS retrieval, schema-aware context, and a dynamic knowledge base
- Applying prompt engineering and incorporating user feedback loop to improve SQL query accuracy and robustness, addressing challenges in multi-table joins and schema alignment
- Enabling cross-cohort analysis and structured reasoning across survey, clinical, and wearable data, with outputs supporting visualizations and ground-truth validation

Undergraduate Researcher – Medical Imaging & Machine Learning

Sept. 2024 – April 2025

MiDATA Lab, Dr. Pascal Tyrrell

Toronto, Canada

- Developed a radiomics-incorporated CNN model in PyTorch to classify adnexal masses utilizing transfer learning on 2,658 ultrasound images from 243 patients
- Achieved a 10% accuracy and 15% AUC improvement over baseline by implementing robust data pipelines for resampling and augmentation to resolve class imbalances
- Validated model performance using 5-fold cross-validation, bootstrap resampling, and noise robustness testing to ensure clinical reliability

Full-Stack Software Engineer Intern

April 2024 – Dec. 2024

Crypto-Arsenal

Taipei, Taiwan (Remote)

- Developed full-stack features for ZKE, a top-100 global cryptocurrency exchange, by building dynamic UI components with TypeScript, React, and Tailwind CSS while optimizing backend operations via Node.js and GraphQL
- Enhanced team productivity through detailed code reviews and active participation in Scrum meetings

Projects

Smart Exam Scheduler (AWS Hackathon) | Javascript, React, AWS SageMaker, S3

Nov. 2024

- Won 1st place for developing a centralized platform to optimize university exam schedules, minimizing room overlaps and student density periods
- Collaborated in a team of 4 to train ML models on SageMaker and manage large-scale datasets via S3 to enable real-time conflict detection

AI Journaling Companion (UofTHacks) | Node.js, Gemini API, Backboard SDK

Jan. 2026

- Designed a multi-model journaling app that uses Gemini for intelligent model routing and Backboard SDK for thread-based context management across weekly user entries
- Integrated Amazon Rekognition for emotion detection and ElevenLabs for TTS to gamify the journaling experience